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History of Astronomy: The Telescope Era

Galileo and His Legacy

Aira Astralis

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Introduction

For all but the last sliver of human history, astronomy was done with unaided eyes. The watchers of Babylon, the geometers of Alexandria, the astrologers of the imperial courts of China, and the navigators of the Indian Ocean all worked within the same narrow band of vision: a pupil about as wide as a pencil, gathering whatever light happened to fall on it. Everything else, every prediction and every calendar, was built on that thin harvest. Then, in the autumn of 1608, a device made of two spectacle lenses arranged in a tube was demonstrated in the Netherlands, and within eighteen months Galileo Galilei had turned such a tube toward the night sky. The harvest multiplied a hundredfold overnight, and it has kept multiplying ever since.

This book is about the four centuries that followed: the telescope era, the period during which the instrument, not the eye, became the primary organ of astronomy. The story runs from a Dutch patent application through Kepler's optics, Newton's mirrors, the giant refractors of the nineteenth century, the mountain observatories of the twentieth, the radio dishes that heard the universe hissing, the segmented mirrors that broke the single-glass ceiling, and the space telescopes that lifted the whole enterprise above the atmosphere. It ends with machines still under construction in Chile, whose primary mirrors will gather a hundred million times the light of the unaided eye.

The telescope did not simply enlarge the sky. It changed what astronomy could ask. With the eye alone, positions and motions were the only measurable quantities; a telescope attached to a graduated instrument made faintness itself a subject of study, and when spectroscopy arrived, the chemical composition of a star became a laboratory question. The universe acquired a history, a chemistry, and eventually a size. None of that was conceivable before the tube of glass.

Two cautions frame the chapters ahead. First, invention is rarely a single event: the telescope's origin is contested among three Dutchmen, and nearly every design that follows has parallel claimants. Second, the machine alone is never the discovery; behind every celebrated instrument stands a chain of craftsmen, budget officers, politicians, and observers, and the honest history of the telescope era is as much about them as about the glass. The chapters that follow keep both cautions in view.

Chapter 1: The Discovery of The Telescope Era

The conventional birth date is October 1608, when Hans Lipperhey, a spectacle maker working in Middelburg in Zeeland, applied to the States General of the Netherlands for a patent on a device "for seeing faraway things as though nearby." The timing is documented; the inventor is not. Within weeks, two rivals appeared: Jacob Metius of Alkmaar filed a competing claim, and a third tradition credits Zacharias Janssen, another Middelburg spectacle maker, whose workshop stood a few streets from Lipperhey's. Modern scholarship cannot settle the priority, and it may not exist to settle: the trick of combining two lenses was probably rediscovered more than once in the lens-grinding towns of the Low Countries around the turn of the century. What is certain is that the States General, persuaded mainly that the device would be useful for spotting ships and armies, declined the patent because others could make the thing, and instead bought several instruments from the applicants. The telescope entered the world as a military toy, with no thought of the sky.

Word moved faster than any ship. By spring 1609, telescopes three powers strong were on sale in Paris, and in Venice that summer, word of the novelty reached a mathematician at Padua who was looking for a way to improve his salary. Galileo Galilei did not invent the telescope, and he never claimed to; he heard a description of it, deduced the arrangement of lenses from optics alone, ground his own, and then did what nobody else had bothered to do. He aimed it upward.

The sky of 1609 was, in the astronomical doctrine of the day, a finished and changeless place. Galileo's tube dismantled it in about two months. The Moon, he found, was not a polished sphere but a landscape, with mountains that cast shadows he could measure by geometry. The Milky Way resolved into uncounted individual stars, more than had ever been credited to the heavens. Four moving points accompanied Jupiter and kept pace with it, a little planetary system that revolved around something other than the Earth, which is what the Copernican arrangement required and the Ptolemaic one struggled to allow. Venus showed phases like the Moon's, meaning it shone by reflected light and circled something that was not the Earth. He published all this in March 1610 in a thin book, the Starry Messenger, and dedicated the moons of Jupiter to the Medici as the "Medicean stars."

There is a quiet complication in the credit story, one the era set the pattern for: Thomas Harriot, an English mathematician financed by the ninth Earl of Northumberland, had a Dutch tube in London by the summer of 1609

and drew the Moon's crescent face through it weeks before Galileo published anything, with a map showing recognizable features. Harriot never published, and his priority languished in his notebooks for two centuries. The lesson was not lost on the discipline that grew from the episode: the era belongs to those who show their work, and a discovery unannounced, in 1609 or today, is a discovery not made.

Confirmation followed quickly, and unevenly. The Senate of Venice could see the moons from the tower of San Marco; philosophers elsewhere were slower, since a good tube was hard to obtain and a bad one showed nothing at all, and the earliest public disputes were fought across exactly that gap in instrument quality. In 1611 the Jesuit astronomers of the Roman College, including Christoph Scheiner, repeated the observations under Clavius's direction and reported them correct, which mattered for reasons diplomatic as much as scientific: the Church's own mathematicians had seen what Galileo claimed. Kepler, in Prague, had asked for a telescope to check Jupiter's moons and got one only after writing the necessary theory with no instrument at all; his 1611 booklet, which named the design later generations would build, was written entirely from first principles by a man who at the time owned no tube.
The Padua senate doubled his pay; the philosophers of Florence did not; the argument over the sky moved from lectures to telescopes.

The instrument that did this was poor by any later standard. Galileo's best tubes magnified about thirty times, showed a field of view the width of the full Moon or less, inverted nothing (his arrangement, a convex objective with a concave eyepiece, showed things upright but very narrow), and choked on color at the edges. Other astronomers confirmed his findings anyway, once they obtained good enough glass; and one correspondent in particular, Johannes Kepler in Prague, both confirmed and improved. Kepler had already published the optical theory showing how the eye focuses; in 1611 he brought out a small treatise on the new instrument in which he proposed putting a convex eyepiece behind a convex objective. The resulting tube inverted the image but had a far wider, brighter field, and by the 1640s, astronomers had learned to live with upside-down skies, because a wider field matters more than an upright one when the object is a planet. The Keplerian design is the ancestor of almost every refractor sold to amateurs today, four hundred years later.

There is a sober epilogue to the discovery chapter. Galileo'scelestial announcements were made over furious resistance; he won the argument over the sky and lost a later argument with the Church over the authority to describe it, spending his last years under house arrest. But the larger point was settled early and permanently: once the telescope showed what a telescope could show, no decree could un-see the phases of Venus. The era had begun. Its next lesson, slower to arrive, was that the first telescopes were crippled by a flaw built into their own glass, and the history of the instrument ever since is largely the history of getting around that flaw.